

Measurement, Ethical Risks, and Data Engineering Requirements

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A Real Case: Healthcare Risk Scores

Goal: Identify patients who need additional care.

- A widely used algorithm predicted **healthcare costs**.
- Costs were used as a proxy for **healthcare needs**.
- At similar risk scores, Black patients were often sicker than White patients.
- The system could overlook patients who needed help.

What went wrong?

The algorithm could predict its chosen target while misrepresenting the underlying need.

Source: Obermeyer et al. (2019), *Science*, DOI: 10.1126/science.aax2342.

Three Ways Data Can Mislead Us

Operationalism Define an abstract concept using observable measurements.

Question: Does the proxy capture what matters?

Reification Treat a constructed score or category as if it were the real-world property.

Question: Are we confusing the score with reality?

McNamara fallacy Make decisions using measurable indicators while discounting what is difficult to quantify.

Question: What important outcomes are we ignoring?

Key distinction: Operational definitions are necessary; misinterpreting or overvaluing them creates risk.

Where the Problem Enters a Data System

Real-world goal $\longrightarrow$ Proxy measure
 $\longrightarrow$ Data pipeline $\longrightarrow$ Model
 $\longrightarrow$ Decision $\longrightarrow$ Human outcome

- **Measure:** Is the target a meaningful proxy?
- **Pipeline:** Who is missing or misrepresented?
- **Model:** Is the score being treated as fact?
- **Decision:** Are other important outcomes ignored?

Engineering principle

Validate the meaning of the data, not only its format.

Data Quality Is Necessary, Not Sufficient

Quality check	What it cannot establish
Complete records	Whether people missing from the dataset matter
Valid values	Whether a valid value describes reality
Consistent joins	Whether linked records belong to the same person
Timely events	Whether the chosen metric captures the desired goal

Example

A resume pipeline can parse every document correctly and still rank applicants using a biased target.

Add: representativeness checks, construct validation, subgroup evaluation, and documentation of uncertainty.

Example 1: Automated Hiring

Goal: Screen resumes efficiently and fairly.

Pipeline: Kafka → Spark → Lakehouse → Ranking service

- **Proxy trap:** Historical hiring decisions stand in for applicant potential.
- **Ethical risk:** Past exclusion can become an apparently objective ranking.
- **Engineering controls:** Dataset lineage, subgroup audits where appropriate, drift alerts, and human reconsideration.

Important: Removing names does not eliminate all bias; other features may act as proxies.

Example 2: Healthcare Allocation

Goal: Identify patients for additional care.

Pipeline: Flink → Secure database → Risk service; lineage tracking

- **Proxy trap:** Cost is substituted for care need.
- **Ethical risk:** Unequal access to healthcare distorts the estimate of need.
- **Engineering controls:** Clinically validated outcomes, subgroup evaluation, data lineage, and documented clinician review.

Important: A clinician's override is a safeguard, not a substitute for fixing the target definition.

Example 3: Smart City Safety

Goal: Improve emergency response and public safety.

Pipeline: MQTT → Time-series database → Quality checks → Dashboard

- **Proxy trap:** Average response time becomes the only success measure.
- **Ethical risk:** Unsafe routing or uneven neighborhood coverage goes unnoticed.
- **Engineering controls:** Response-time distributions, collision rates, geographic coverage, and community feedback.

Important: Hypothetical design scenario; faster response does not itself imply less safety.

Example 4: Context-Aware Moderation

Goal: Identify potentially harmful content.

Pipeline: APIs → Text index → NLP scoring → Review workflow

- **Proxy trap:** A toxicity score stands in for the meaning of a message.
- **Ethical risk:** Satire, quotation, and language varieties are misclassified.
- **Engineering controls:** Review for uncertain cases, context retention, appeals, and evaluation across language varieties.

Important: Dead-letter queues handle technical failures; ambiguous content needs a human-review queue.

Capstone Matrix: Architecture vs. Ethics

Project topic	Primary system requirement	Ethical failure mode	Engineering solution
Hiring	Scalable intake and traceability	Past bias becomes applicant ranking	Lineage, subgroup audits, review
Healthcare	Low latency, privacy, provenance	Costs mistaken for care needs	Clinical targets, validation, review
Smart city	Sensor scale and data quality	Response-time fixation	Safety metrics, geographic checks
Moderation	Text indexing and orchestration	Context-blind removals	Uncertainty routing, appeals

Common theme: Ethical safeguards are system requirements, not optional post-processing.

Turn Ethical Risks into Requirements

1. **Define the goal:** What human outcome matters?
2. **Justify the proxy:** What does the metric miss?
3. **Test the data:** Who is absent or misrepresented?
4. **Monitor harm:** Which counter-metrics matter?
5. **Enable review:** Who can correct a decision?
6. **Preserve evidence:** Can we trace and audit it?

Design test

Can the system detect, explain, and correct a decision that is technically valid but substantively wrong?

Conclusion

Three lessons

- A **proxy** is not the underlying reality.
- An accurate **model** need not serve the right goal.
- A reliable **pipeline** need not produce fair decisions.

Discussion: A pipeline passes every data quality test and its model achieves 95% accuracy. What else must we know before trusting its decisions?

Further reading:

- Obermeyer et al. (2019), *Science* Vol. 366, No. 6464,
DOI: 10.1126/science.aax2342
<https://www.science.org/doi/10.1126/science.aax2342>
- NIST AI Risk Management Framework
<https://www.nist.gov/itl/ai-risk-management-framework>
- Wikipedia: Reification (fallacy)
[https://en.wikipedia.org/wiki/Reification_\(fallacy\)](https://en.wikipedia.org/wiki/Reification_(fallacy))
- Wikipedia: McNamara fallacy
https://en.wikipedia.org/wiki/McNamara_fallacy